6.4 (a) From Eq. (5.105), the magnetic induction inside the sphere is $\vec{B} = \frac{2}{3}\mu_0 \vec{M}$. Due to the rotation of the sphere, electric field is generated, which is given by $\vec{E} = \vec{v}\times\vec{B}$, where $\vec{v} = -\vec{\omega}\times\vec{r}$.

Therefore, $\vec{E} = -\frac{2}{3}\mu_0(\vec{\omega}\times\vec{r})\times\vec{M} = -\frac{2}{3}\mu_0 M\omega r\left[(\hat{z}\times\hat{r})\times\hat{z}\right] = -\frac{2}{3}\mu_0 M\omega \vec{r}_\perp$,

where $\hat{r} = \hat{z} + \hat{r}_\perp$. Then, the induced charge density is

$$\rho = \varepsilon_0 \nabla\cdot\vec{E} = -\frac{2}{3}\mu_0\varepsilon_0 M\omega\,\nabla\cdot\vec{r}_\perp = -\frac{4\omega M}{3c^2}.$$

Since $M = \frac{3m}{4\pi R^3}$, we can also write the charge density as $\rho = -\frac{m\omega}{\pi c^2 R^3}$.

(b) The scalar potential outside the sphere satisfies the Laplace equation and can be expressed as $\Phi_{out} = \sum_{l=0}^{\infty}\frac{B_l}{r^{l+1}}P_l(\cos\theta)$. Then, the electric field is given by

$$\vec{E}_{out} = -\nabla\Phi_{out} = -\hat{r}\frac{\partial\Phi_{out}}{\partial r} - \hat{\theta}\frac{1}{r}\frac{\partial\Phi_{out}}{\partial\theta} = \hat{r}\sum_{l=0}^{\infty}\frac{(l+1)B_l}{r^{l+2}}P_l(\cos\theta) - \hat{\theta}\sum_{l=0}^{\infty}\frac{B_l}{r^{l+2}}P_l'(\cos\theta),$$

where we have used the fact that $dP_l(\cos\theta)/d\theta = P_l'(\cos\theta)$. But the inside electric field is

$$\vec{E}_{in} = -\frac{2}{3}\mu_0 M\omega r\sin\theta\,\hat{r}_\perp = -\frac{2}{3}\mu_0 M\omega r\sin\theta\left(\hat{r}\sin\theta + \hat{\theta}\cos\theta\right).$$

By the continuity of tangential electric field at the surface, we must have $B_l = 0$, $l \neq 2$, and

$$-\frac{2}{3}\mu_0 M\omega R\sin\theta\cos\theta = -\frac{B_2}{R^4}(-3\sin\theta\cos\theta), \quad \text{or} \quad B_2 = -\frac{2}{9}\mu_0 M\omega R^5$$

Comparing with the expansion of scalar potential in spherical harmonics, Eq. (4.1),

$$\Phi(\vec{x}) = \frac{1}{4\pi\varepsilon_0}\sum_{l=0}^{\infty}\sum_{m=-l}^{l}\frac{4\pi}{2l+1}q_{lm}\frac{Y_{lm}(\theta,\phi)}{r^{l+1}},$$

only $l=2$, $m=0$ is present. Then,

$$\Phi(\vec{x}) = \frac{1}{5\varepsilon_0}q_{20}\frac{Y_{20}(\theta,\phi)}{r^3} = \frac{1}{5\varepsilon_0}\sqrt{\frac{5}{4\pi}}\,q_{20}\frac{P_2(\cos\theta)}{r^3} = B_2\frac{P_2(\cos\theta)}{r^3},$$

and

$$q_{20} = \sqrt{\frac{4\pi}{5}}\,5\varepsilon_0 B_2$$

From Eq. (4.6), we know $Q_{33} = 2\sqrt{\frac{4\pi}{5}}\,q_{20} = 8\pi\varepsilon_0 B_2$. The tracelessness of the quadruple moment tensor implies that $Q_{11} = Q_{22} = -\frac{1}{2}Q_{33}$.

Therefore, we have found all quadrupole moments, with

$$Q_{33} = 8\pi \varepsilon_0 B_2 = -\frac{16}{9} \mu_0 \varepsilon_0 \pi M \omega R^5 = -\frac{16}{9c^2} \pi \omega R^5 \frac{m}{\frac{4}{3}\pi R^3} = -\frac{4 m \omega R^2}{3c^2}$$

(c) The charge density at the surface is given by

$$\sigma(\theta) = \varepsilon_0 \left(E_{out,r} - E_{in,r} \right)\Big|_{r=R} = \varepsilon_0 \left(\frac{3B_2}{R^4} P_2(\cos\theta) + \frac{2}{3} \mu_0 M \omega R \sin^2\theta \right)$$

Since $\sin^2\theta = \frac{2}{3}\left(1 - P_2(\cos\theta)\right)$, we then have

$$\sigma(\theta) = \varepsilon_0 \left(-\frac{2}{3} \mu_0 M \omega R \, P_2(\cos\theta) + \frac{2}{3} \mu_0 M \omega R \cdot \frac{2}{3}\left(1 - P_2(\cos\theta)\right)\right)$$

$$= \varepsilon_0 \left(\frac{4}{9} \mu_0 M \omega R - \frac{10}{9} \mu_0 M \omega R \, P_2(\cos\theta) \right) = \frac{4}{9} \mu_0 \varepsilon_0 M \omega R \left(1 - \frac{5}{2} P_2(\cos\theta) \right)$$

$$= \frac{4 \omega R}{9c^2} \frac{m}{\frac{4}{3}\pi R^3} \left[1 - \frac{5}{2} P_2(\cos\theta) \right] = \frac{m\omega}{3\pi c^2 R^2} \left[1 - \frac{5}{2} P_2(\cos\theta) \right]$$

$$= \frac{1}{4\pi R^2} \frac{4 m \omega}{3 c^2} \cdot \left[1 - \frac{5}{2} P_2(\cos\theta) \right]$$

(d) The potential difference from the equator to the pole can be determined by

$$\mathcal{E} = -\int_0^R \vec{E} \cdot d\vec{r}_\perp = \frac{2}{3} \mu_0 M \omega \int_0^R r_\perp \, dr_\perp = \frac{1}{3} \mu_0 M \omega R^2$$

$$= \frac{1}{3} \mu_0 \omega R^2 \cdot \frac{m}{\frac{4}{3}\pi R^3} = \frac{\mu_0 m \omega}{4\pi R}$$